

Embedded Systems Course

6 Months Professional Training Program | 70% Practical + 30% Theory

Course Overview

Duration: 6 Months

Approach: 70% Practical + 30% Theory

Languages: Embedded C / C++

Microcontrollers: 8051, Arduino, PIC (PIC16F877A)

Tools: Keil, Proteus, Arduino IDE, Wokwi, MPLAB X, XC8

Month 1 — Embedded C & 8051 Fundamentals

Topics Covered

- Introduction to Embedded Systems
- Microcontroller vs Microprocessor
- Embedded C Fundamentals
- Variables, Data Types and Operators
- Conditional Statements and Loops
- Functions, Arrays and Strings
- Pointers and Structures
- Bitwise Operations
- Registers and Memory Concepts
- 8051 Architecture and I/O Ports

Practical Labs

- LED Blinking and LED Patterns
- Push Button and Buzzer Interfacing
- 7-Segment Display Interfacing
- Switch-Based Applications

Tools: Keil + Proteus + 8051 Development Board

Month 2 — 8051 Interfacing & Peripherals

Topics Covered

- LCD and Keypad Interfacing
- Timers and Counters
- Interrupts
- UART / Serial Communication
- ADC and PWM Fundamentals
- Sensor Interfacing

Mini Projects

- Digital Counter and Digital Clock
- Password-Based Security System
- Temperature Monitoring System
- Serial Communication Application

Tools: Keil + Proteus + 8051 Hardware

Month 3 — Arduino Fundamentals & Hardware Interfacing

Topics Covered

- Arduino Architecture and Arduino Uno
- Arduino IDE, Digital and Analog I/O
- Serial Communication and PWM
- Functions, Libraries, Sensors and Actuators

Practical Labs

- LEDs, RGB LEDs, Push Buttons, Potentiometers
- LDR, Temperature Sensor, IR Sensor
- Buzzer, Servo Motor, DC Motor, Relay
- Ultrasonic Sensor

Mini Projects

- Automatic Street Light
- Obstacle Detection System
- Temperature Monitoring System

Tools: Arduino Uno + Arduino IDE + Wokwi

Month 4 — Communication Protocols & Arduino Projects

Communication Protocols

- **UART:** TX/RX, Baud Rate, Data Transmission and Reception
- **I2C:** Master-Slave Concept, SDA/SCL, Addressing, Sensor/Display Interfacing
- **SPI:** MOSI, MISO, SCLK, CS/SS, Master-Slave Communication
- **CAN Basics:** Overview, Bus Architecture, Nodes and Messages, Automotive Applications

Topics Covered

- I2C LCD/OLED and I2C Sensor Interfacing
- SPI Device Interfacing and UART Communication
- Multi-Device Communication

Mini Projects

- Smart Home Automation
- Sensor Monitoring System
- Serial-Controlled Device

Tools: Arduino IDE + Hardware Modules

Month 5 — PIC Microcontroller

Topics Covered

- PIC Architecture (PIC16F877A)
- GPIO, Registers, ADC, Timers, Interrupts
- PWM, LCD, Keypad, UART, Sensor Interfacing

Practical Labs

- LED, Switch and LCD Interfacing
- ADC-Based Sensor Reading and Temperature Monitoring
- Motor Control, UART Communication and PWM Applications

Tools: MPLAB X IDE + XC8 + Proteus + PIC Development Board

Month 6 — Advanced Concepts & Final Project

Topics Covered

- Inter-Microcontroller Communication (UART / I2C / SPI)
- CAN Fundamentals and Interrupt-Driven Programming
- Timers, PWM, ADC and Sensor Integration
- Debugging Techniques and Datasheet Reading
- Basics of Embedded System Design
- Introduction to RTOS Concepts
- Git/GitHub Basics

Tools: All Previous Tools + Git/GitHub

Final Project

Students will develop a complete embedded system integrating microcontroller, Embedded C, sensors, actuators, display, communication protocols, and interrupts/timers.

Suggested Projects

- Smart Home Automation System
- IoT-Based Temperature Monitoring System
- Smart Parking System
- Obstacle Detection and Alert System
- Automated Street Light System
- Industrial Temperature and Safety Monitoring System

Assessments

- **Weekly Test:** Conducted at the end of every week based on topics covered.
- **Weekly Assignment:** Embedded C programs, circuit simulations, microcontroller programming, sensor interfacing, debugging exercises, protocol-based tasks, and mini-projects.

Eligibility

- Diploma Students
- BE / B.Tech Students
- ME / M.Tech Students
- BSc / MSc in Electronics

Contact & Enrollment

Institute Name: Embedded Electronics

Website: <https://embedded-electronics.github.io/>

Email: embeddedelectronics.in@gmail.com

Start your journey into Embedded Systems today.

Learn online. Build real projects. Become industry-ready.